



ICERM

Summer 2024 Newsletter



BROWN

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Updates from Director Brendan Hassett

Greetings!

This month, ICERM will wrap up another event-filled summer. Continuing our tradition of filling the summer months with as much math as possible, ICERM has welcomed a different workshop or program almost every week since the beginning of May. We do our best to touch on as many areas of mathematics and computation as possible based on recommendations from the global mathematical research community. This summer, our programs spanned topics as diverse as simulating extreme spacetimes, the topology of hypergraphs, and spectral analysis of Schrödinger operators.

Summer is also the time we welcome budding mathematicians for [GirlsGetMath@ICERM](#) and Summer@ICERM, our annual programs for high school and undergraduate students to develop and hone their mathematical research skills. Each year, these programs allow faculty to guide the next generation of mathematical thinkers in hands-on projects designed to challenge them and introduce stimulating scientific questions. This year's Summer@ICERM program [Mathematical Models to Predict, Prepare, and Prevent](#) invited undergraduates to create models to answer questions in epidemiology and sports analytics. You may be able to catch some of our recent alumni at the 2025 Joint Mathematics Meetings!

As the summer comes to an end, I am eagerly looking forward to ICERM's upcoming Fall Semester Program [Theory, Methods, and Applications of Quantitative Phylogenomics](#). Participating researchers will use their mathematics, statistics, computer science, and biology expertise to bridge new knowledge about interspecies interactions with ever-increasing available genomic data. We expect their collaborations to help shape the field of phylogenetics for years to come, and we won't be surprised if visiting mathematicians and computationalists develop some novel methods in the process! If you are a biologist, computer scientist, or mathematician interested in biology, we invite you to apply to the [October](#) and [November](#) Semester Workshops.

This will be the first Semester Program since ICERM Deputy Director Caroline Klivans completed her term in June. I am grateful for her insightful leadership and scientific vision, which has benefitted ICERM in innumerable ways since her appointment as Associate Director in 2015. We look forward to her return in 2025, when she will rejoin ICERM as a member of the PI team and an advisor on programs related to artificial intelligence and computer science.

I am excited to announce that Nathaniel Whitaker will join ICERM as Associate Director for the 2024-2025 academic year! He joins us after retiring from the University of Massachusetts Amherst, where he was chair of the Math & Statistics department and interim dean of the College of Natural Sciences. Before joining ICERM's team of directors, Whitaker was a regular organizer and participant in workshops related to [mathematical biology](#) and fluid mechanics, as well as an [ICERM public lecturer](#)! As Associate Director, he will help supervise ICERM's scientific programs and shape the long-term vision for the Institute.

There have also been transitions within ICERM's Board of Trustees and Advisory Boards. I want to extend my heartfelt thanks to Ron Buckmire, Anna Gilbert, and Leslie Greengard of ICERM's Board of Trustees; Jennifer Balakrishnan, Michael Holst, Yael Kalai, Daniel Krashen, and Nathan Kutz of the Scientific Advisory Board; and Katharine Ott of the Educational Advisory Board for their years of service. Their perspectives have been invaluable to ICERM leadership and staff as we've worked to support an ambitious array of cutting-edge programs, mathematical educational initiatives, and collaborations. You can learn more about ICERM governance and our twelve new trustees and advisory board members at [icerm.brown.edu/about/boards/](#).

Finally, I'd like to take a moment to remember [Jim Simons](#), who passed away in May. Simons was a singularly gifted mathematician and businessman, and founded several philanthropic organizations dedicated to advancing scientific progress. His work inspired many and shaped the trajectory of mathematics research for decades to come, including at ICERM. Thanks to the generosity of the Simons Foundation and Math for America, two organizations founded by the Simons family, ICERM has been able to support two programs at the core of our institutional identity: the [Simons Collaboration on Arithmetic Geometry, Number Theory, and Computation](#) and GirlsGetMath@ICERM. I look forward to continuing these programs and am committed to carrying on Simons' legacy of shaping the world through mathematical excellence in all we do here at ICERM.

We look forward to seeing you at your next program!

Sincerely,



Brendan Hassett

UPCOMING OPPORTUNITIES

Call for proposals: Summer@ICERM 2025

ICERM is seeking proposals for Summer@ICERM 2025, an eight-week summer research experience for undergraduates. Summer@ICERM is led by faculty organizers who develop a variety of interdisciplinary research projects in an area of their choosing. Summer@ICERM is not your typical REU, in that ICERM provides comprehensive start-to-finish logistical support to the faculty organizers, including assistance recruiting TAs. Faculty and TAs are paid summer stipends and salaries with lodging support. Accepted undergraduates receive stipends and funding for room, board, and travel to Providence, RI. Proposals received by the first week of September will receive full consideration. Prospective organizers should contact the [ICERM Director](#) prior to submitting a formal proposal.

Apply to attend the 2024 Blackwell-Tapia Conference at ICERM

The conference and prize honor Dr. David Blackwell, the first African-American member of the National Academy of Science, and Dr. Richard Tapia, winner of the National Medal of Science in 2010, two seminal figures who have inspired generations of Black, Latinx, and Indigenous students and researchers in the mathematical sciences.

NSF Partnerships for Research Innovation in the Mathematical Sciences Award

The PRIMES program aims to support partnerships between [NSF Division of Mathematical Sciences-supported institutes](#) and faculty at [minority-serving institutions](#). The program seeks to boost the participation of members of underrepresented groups in the mathematical sciences through their increased involvement in research programs at the institutes. The next deadline for proposals is **February 12, 2025**.

Modern Mathematics Workshop 2024

Attend the Modern Mathematics Workshop (October 30-31) at the 2024 NDiSTEM Conference of the Society for the Advancement of Chicanos and Native Americans in Science (SACNAS). Funding is available to cover travel and lodging and SACNAS NDiSTEM registration for participants in Modern

Math, as well as to fund lodging for workshop participants who also wish to attend the NDiSTEM Conference. The workshop and conference will take place in Phoenix, Arizona.

Latinx in the Mathematical Sciences Conference (LatMath) 2025

The Institute for Pure and Applied Mathematics (IPAM) will host a conference showcasing the achievements of Latinx mathematicians. The goals of the conference are to encourage Latinx students to pursue careers in the mathematical sciences, to promote the advancement of Latinx researchers currently in the discipline, to showcase research being conducted by Latinx mathematicians at the forefront of their fields, and to build a community around shared academic interests. The conference will be held on the UCLA campus in Los Angeles, CA.

Want to share an opportunity in ICERM's next newsletter? Submit your program or event to communications@icerm.brown.edu.

UPCOMING PROGRAMS

Semester Programs

Fall 2024: [Theory, Methods, and Applications of Quantitative Phylogenomics](#)

- Sep 16 - 20: Current Methods and Open Problems in Mathematical and Statistical Phylogenetics
- Oct 21 - 25: From Phylogenetics to Phylogenomics: Mathematical and Statistical Challenges in the Era of Big Data
- Nov 18 - 22: Algorithmic Advances and Implementation Challenges: Developing Practical Tools for Phylogenetic Inference

Spring 2025: [Geometry of Materials, Packings, and Rigid Frameworks](#)

- Feb 10 - 12: Circle Packings, Minimal Surfaces, and Discrete Differential Geometry
- Mar 17 - 21: Matroids, Rigidity, and Algebraic Statistics
- Apr 7 - 11: Geometry of Materials

Fall 2025: [Categorification and Computation in Algebraic Combinatorics](#)

Spring 2026: [Stochastic and Randomized Algorithms in Scientific Computing: Foundations and Applications](#)

Topical Workshops

Dec 9 - 13, 2024: [Harmonic Analysis and Convexity](#)

Jan 6 - 10, 2025: [Computational Learning for Model Reduction](#)

Jan 13 - 17, 2025: [Women in Mathematical Computational Biology](#)

Jan 21 - 24, 2025: [Patterns, Dynamics, and Data in Complex Systems](#)

May 12 - 16, 2025: [Links in Dimensions 3 and 4](#)

Jun 2 - 6, 2025: [Scientific Machine Learning for Gravitational Wave Astronomy](#)

Jun 23 - 27, 2025: [Algebraic Points on Curves](#)

Jul 7 - 11, 2025: [LMFDB, Computation, and Number Theory \(LuCaNT\) 2025](#)

Jul 21 - 25, 2025: [Innovative and Efficient Strategies for Stiff Differential Equations](#)

Aug 11 - 15, 2025: [Illustrating Math: Reunion & Expansion](#)

PROPOSE A PROGRAM

ICERM is accepting proposals for the following program types:

Semester Programs

ICERM hosts a Spring and Fall Semester Program every year. Each has organizers and long-term participants who are expected to be in-residence for the majority of the semester. Semester programs typically incorporate three week-long associated workshops. For a typical program, ICERM provides travel and lodging support for 5-10 organizers (at least a month in residence, with some for the full semester); 20-30 long-term participants (4+ weeks); 1 Institute and 5 Semester postdoctoral fellows (salaried); 60 short-term participants (1-4 weeks); 10-15 graduate students (6+ weeks); and workshop attendees. ICERM helps essential long-term participants negotiate sabbatical leaves or teaching releases to foster their participation.

Summer@ICERM

Summer@ICERM is a Research Experience for Undergraduates led by faculty organizers who develop a variety of topical research projects that are interdisciplinary in nature. Throughout the eight-week program, students work on their assigned projects in groups of two-to-four, mentored by faculty advisors and teaching assistants. Summer@ICERM is not your typical REU, in that ICERM provides comprehensive start-to-finish logistical support to the faculty organizers, including assistance recruiting TAs. In addition, faculty and TAs are paid summer salary with lodging support, and the average faculty/TA-to-student ratio is 1:3.

Topical Workshops

Topical Workshops run over 5 weekdays and focus on a topical theme that aligns with the Institute’s mission of supporting and broadening the relationship between mathematics and computation. ICERM hosts 5-7 topical workshops each year. They are typically scheduled in December, January, and May through August (around the dates of the semester programs). ICERM covers travel and lodging expenses of the organizers and 20-25 invited speakers/lead participants; some participants (applicants) to the workshop may also be supported.

Graduate Training Workshops

Graduate Training Workshops are 1-3 week events on a timely and exciting mathematical or computational theme, with a focus on training graduate students, recent PhDs, and researchers new to the field. They are typically scheduled in December, January, and May through August (around the dates of the semester programs). ICERM covers travel and lodging expenses of the organizers, invited speakers, TAs, and early-career participants.

Collaborate@ICERM

Collaborate@ICERM offers teams of 3-6 researchers the opportunity to spend five days at the Institute during the summer (May-August) or during the month of January. The team research project should have a computational or experimental component. ICERM provides access to a variety of software packages and high performance computing through Brown’s [Center for Computation and Visualization](#). Proposals involving research projects initiated at past ICERM programs are encouraged. This program provides limited funding for travel to the Institute and local accommodations for six nights.

Hot Topics Workshops

Hot Topics Workshops are designed to allow ICERM to move quickly in order to start the public exploration of breakthroughs and emerging mathematical areas. They run 2-3 days and are organized on a few months’ notice. They can originate through suggestions from ICERM boards or the community.

Deadlines and Review Process

Hot Topics Workshop suggestions are reviewed on a rolling basis. The next target deadline for all other proposal types is **October 1, 2024**. Submissions and inquiries should be sent to [ICERM’s Director](#). ICERM’s Directorate and the Scientific Advisory Board (SAB) review all proposals. Proposers will receive feedback within a month of the November SAB meeting.

For more information, visit icerm.brown.edu/proposals.

Research Spotlight: Marissa Masden

Marissa Masden was an Institute Postdoctoral Fellow at ICERM during the Fall 2023 [Math + Neuroscience Semester Program](#) and the Spring 2024 semester. This research highlight is based on a collaboration between Masden and ICERM Semester Postdoctoral Fellow [Robyn Brooks](#) initiated during the Fall 2023 program.

Exploiting combinatorial regularity for topological and Morse measures of ReLU neural networks

Marissa Masden

Based on joint work with Robyn Brooks

The progress of modern machine learning accelerated following the introduction of the *Rectified Linear Unit*, or ReLU, an extremely simplified model of a neuron which has a $\max\{0, x\}$ “activation” function [1]. For the purposes of this highlight, a *ReLU layer* is a function $F_i : \mathbb{R}^{n_{i-1}} \rightarrow \mathbb{R}^{n_i}$ of the form $\mathbf{ReLU} \circ A_i$, where the ReLU $\max\{0, x\}$ is applied coordinatewise after an affine function A_i . A *(fully-connected, feedforward) ReLU neural network* is a function composed of ReLU layers of the form:

$$F : \mathbb{R}^{n_0} \xrightarrow{F_1} \mathbb{R}^{n_1} \xrightarrow{F_2} \dots \xrightarrow{F_m} \mathbb{R}^m \xrightarrow{G} \mathbb{R}$$

where G is an affine function. In machine learning language, the tuple $(n_0, \dots, n_m)$ may be thought of as the *architecture* of the *network* F . A network *learns* when the *parameters* of the affine functions A_i are updated in *training steps* via gradient descent, to make F more closely approximate a target function. Generally, the *layer widths* n_i are fixed throughout this process. ReLU neural networks’ advantages include a computationally inexpensive learning rule for updating the A_i , and no gradient-saturation problem like “traditional” sigmoid models [1].

Identifying the capabilities of a machine learning model before applying it is a key goal in *neural architecture search* [4]. Too small of a model, and the network will never successfully perform the task it is intended for. Too large of a model, and computational energy is wasted and the cost of model training may become unbearably high. While universal approximation theorems [2] and even dynamics [3] are available for various limiting forms of ReLU neural networks, there are still many questions about the limitations of what *small* neural networks of this form are capable of accomplishing: for example, what functions may be expressed when the n_i are fixed and finite? What are the attainable sublevel set topologies when restricting the architecture in this way? Answering these questions can give insight into how many parameters a neural network *must* have before it is capable of performing a particular task.

Marissa Masden seeks to understand ReLU neural networks through a combinatorial, discrete structure which enables capturing topological invariants of structures like the *decision boundary*, the $F(x) = 0$ level set—one way to characterize the classification capability of a network model. Like hyperplane arrangements, ReLU neural networks subdivide their domain $\mathbb{R}^{n_0}$ into a polyhedral complex. Using language from Grigsby and Lindsey, this is called the *canonical polyhedral complex* corresponding to a given ReLU neural network [5], and it can be thought of as an

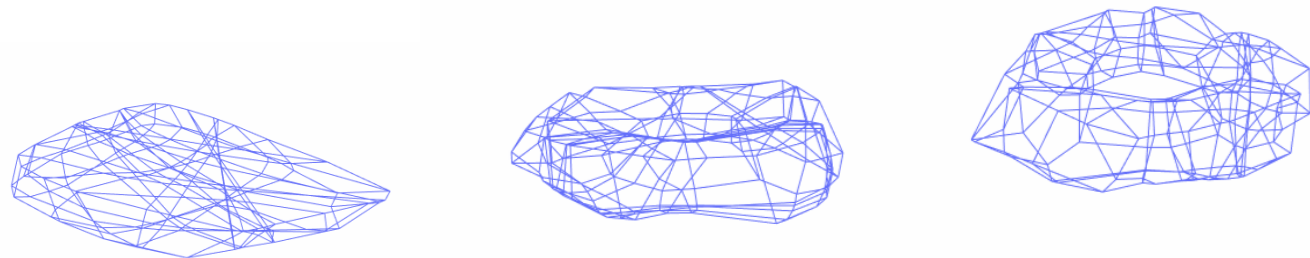


Figure 1: A single bent hyperplane representing the decision boundary of a ReLU neural network at several different training steps as it becomes a topological torus. An interactive version of this graphic, including sign vectors on mouseover, is available at https://mmasden.github.io/assets/embeds/torus_graph.html.

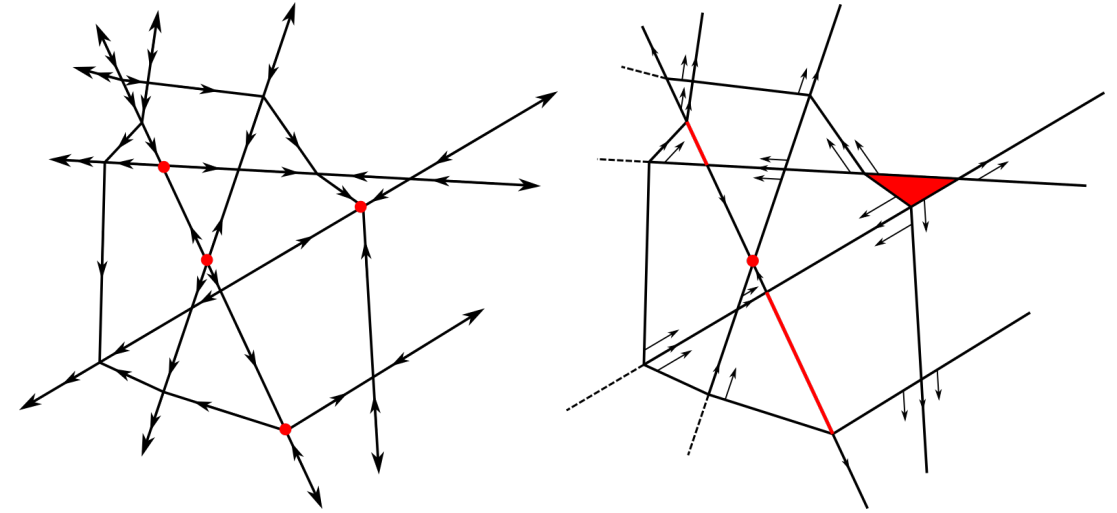


Figure 2: A discrete gradient vector field for the bounded-above cells of the canonical polyhedral complex of a ReLU neural network, realizing a bijective correspondence between PL Morse critical points of index i and Discrete Morse critical cells of dimension i . The locality of the algorithm is emphasized through placement of the arrows indicating discrete Morse coface-face pairs.

intersection arrangement of a set of (generally) codimension-1 *bent hyperplanes*, zero-sets corresponding to individual layers’ affine functions.

As long as sufficient transversality and genericity conditions are met for the layers, the “bent hyperplanes” corresponding to a ReLU neural network have *local* n_0 -fold intersections which are characteristic of hyperplanes in general position [6]. However, *globally* each “bent hyperplane” may have seemingly arbitrary combinatorial and topological type, as demonstrated by Figure 1. Despite this, sign vectors in $\{-1, 0, 1\}^{\sum n_i}$ labeling polyhedra with a sign relative to each bent hyperplane—analogue to the covectors of a hyperplane arrangement oriented matroid—are still sufficient to encode the face poset of the polyhedral complex [6]. Furthermore, the sign vectors of only the vertices of the canonical polyhedral complex may be used to reconstruct the connectivity of the entire polyhedral complex.

This combinatorial structure enables the explicit computation of the decision boundary of a classification neural network, the Betti numbers of the decision boundary, the network’s sublevel and superlevel sets, and the (piecewise linear Morse) critical points of the function [7]. However, due to the number of vertices growing exponentially in the input dimension n_0 , computing the entire polyhedral complex, even only needing vertices and sign vectors, is impractical for networks with even moderately many inputs.

Marissa Masden and Robyn Brooks have developed an local, constructive algorithm for translating between PL Morse functions [8] and Discrete Morse gradient vector fields [9] for ReLU neural networks (depicted in Figure 2), in a bid to identify possible computational and theoretical shortcuts for computing topological expressiveness. In general, such translations may not always be decidable, but in the context of ReLU neural networks the local coordinate-axis-like combinatorial structure at each vertex makes the identification of PL critical points, and a translation to a discrete Morse gradient vector field, tractable. Their preprint is forthcoming.

Marissa Masden was an Institute Postdoctoral Fellow at ICERM during the Semester on Math and Neuroscience and the following semester. She spent her time there working on this problem, extensions of the described computational tools to alternative neural network architectures, and other projects which began during the Semester on Math and Neuroscience.

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[5] J. Elisenda Grigsby and Kathryn Lindsey. On transversality of bent hyperplane arrangements and the topological expressiveness of relu neural networks. *SIAM Journal on Applied Algebra and Geometry*, 6(2):216–242, 2022.

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[7] J. Elisenda Grigsby, Kathryn Lindsey, and Marissa Masden. Local and global topological complexity measures of relu neural network functions, 2022. <https://arxiv.org/abs/2204.06062>.

[8] Romain Grunert. *Piecewise Linear Morse Theory*. PhD thesis, 2017.

[9] Robin Forman. A user’s guide to discrete Morse theory. *Sém. Lothar. Combin.*, 48:Art. B48c, 35, 2002.

This video shows an interactive graphic built by Masden. The figure depicts a bent hyperplane representing the decision boundary of a ReLU neural network at a given training step as it becomes a topological torus. [Try it yourself!](#)

Introducing Blackwell-Tapia Prize Recipient Illya Hicks



ICERM and the National Blackwell-Tapia Prize Selection Committee are pleased to announce the 2024 Blackwell-Tapia Prize will be awarded to [Dr. Illya Hicks](#), Professor and Department Chair of Computational Applied Mathematics and Operations Research at Rice University.

Since 2002, the biennial Blackwell-Tapia Prize has recognized a mathematical scientist who has made outstanding contributions to research in their field of expertise and served as a role model for mathematical scientists and students from underrepresented groups or has contributed in significant ways to addressing the underrepresentation of minorities in math. The prize honors the legacies of the renowned Drs. David H. Blackwell and Richard A. Tapia, whose work has inspired generations of Black, Latinx, and Indigenous students and researchers in the mathematical sciences.

Hicks is a leading researcher of combinatorial optimization, graph theory, and integer programming whose contributions have had a major lasting impact on these fields. He is well-known for his work on the Robertson-Seymour graph decomposition theory, the development of techniques to uncover cohesive structure in sparse graphs, and the computational study of the zero-forcing problem to understand the propagation of information in networks.

In June, ICERM asked Hicks about his current research, his mathematical path, and what he's looking forward to at the November 2024 [Blackwell-Tapia Conference and Award Ceremony](#):

Transcript has been edited for clarity.

ICERM: What do you find most exciting about what you’re currently researching?

Hicks: I actually just got out of a meeting with my student and the number of possibilities with our research is pretty good. This particular piece of research is using combinatorial optimization techniques for optimal control-type things. It’s exciting because of the novelty of it but then also the mixture of the areas and the number of applications you can do with it...It has a lot of different applications.

ICERM: Can you name some of them?

Hicks: Machine learning, optimal control-type problems, so robotics, path-planning for robotics, optimal movement of robotic arms. There was some stuff related to marketing applications, so it has a number of applications. Probably things related to quantum too, but I don’t know that for sure. I definitely know power systems, electrical engineering, and chemical engineering, too. It’s basically a mixed-integer problem with non-linear type constraints.

ICERM: What first inspired you to become a mathematician?

Hicks: When I was younger, I kind of wanted to design cars. My dad was a mechanic, so I guess technically I would’ve become more of a mechanical engineer. But I had a football scholarship and [my school] didn’t have engineering, so I said, “Okay, what can I do? What is engineering?” Well, to me, engineering is math and physics. So I majored in math and minored in physics, and then I took a course in operations research and it kind of interested me. I said, “Oh—I can use math to help companies and solve problems related to industry,” and that kind of piqued my interest. And when I came to Rice and

saw things related to graphs and networks, I said, “Oh—that kind of speaks to my visual preference.” Graphs are kind of like the mathematics of relationships. They show how things are related or unrelated and they satisfy that visual desire I had, but they also have math which I was always pretty good at. When I was younger, I never wanted the test to beat me, so I would kind of be competitive with the test...Sometimes you’re born with innate skills so there are some things you’re more inclined to do, but you can do anything if you’re willing to do it...You have to know what it takes to get to a certain level and you have to be willing to actually work to get [there].

ICERM: As a mentor to many students and early-career mathematicians, what advice do you typically give to up-and-coming researchers?

Hicks: I usually say research requires two things: creativity and perseverance. So you’ve got to be creative, you’ve got to create new knowledge and expand the knowledge base, but you also have to persevere in the trials and tribulations that you may encounter. Another thing is to always try to be visible. Especially if we’re talking about getting tenure, that’s probably one of the hidden criteria, just remaining visible and relating your research. You have to go to conferences, you have to be engaged, you have to display your research and convey your ideas to people. Because you can have the next discovery but if you don’t know how to convey it, it will never be known, right? Also, do things that you enjoy. I tell my own graduate students, “I’d rather you do something that you like or enjoy than be trying to force something down your throat because you’re going to put more effort into things you like to do.” Even with their research topics, I try to find something where I have some expertise or some intersection that’s also something they’re interested in.

ICERM: What are you most looking forward to about this year’s Blackwell-Tapia Conference?

Hicks: Just colleagues, seeing them...It’s always a special thing to be recognized by people you admire. It’s kind of like a family reunion...And then maybe meeting new people. Hopefully, the community is growing and I’ll meet those people as well.

I’m just so humbled to receive the [Blackwell-Tapia Prize]. I appreciate the award but understand that there clearly could be other people in this particular place, so I don’t take it for granted...It’s incredibly humbling to receive the award, and I want to recognize everyone who puts in the effort to help this general cause of increasing diversity, especially when diversity is under attack...We just have to persevere and come up with new ways to do what we do because they can’t stop the show. We’ve done it for so many years and we’ll continue to do it. ■

Learn more and register for Blackwell-Tapia 2024 at icerm.brown.edu/events/btc2024/.

MAKE A GIFT TO ICERM

Through our Semester Programs, Topical and Hot Topics Workshops, Collaborate@ICERM awards, Summer@ICERM undergraduate research programs, and GirlsGetMath@ICERM, we bring researchers at all career stages to Providence, from around the world.

Individual donations to ICERM help us maintain a fund specifically dedicated to meeting the needs of every visitor. This fund enables us to support mathematicians with families, provide assistance to visitors with emergency expenses, and offer free and low-cost community-engaged programs such as GirlsGetMath@ICERM.

[MAKE A GIFT TODAY](#)

Visiting ICERM as a Graduate Student Researcher

by [Han Le](#) and [Alexander Van Werde](#)

Graduate students Han Le (University of Michigan) and Alexander Van Werde (Eindhoven University of Technology) both attended the ICERM Topical Workshop [Random Matrices and Applications](#) in May. They both study random matrices, albeit from different angles – Han explores their connections to statistical physics, while Alexander is interested in their applications to community detection and spectral graph theory.

As early-career researchers, the feeling of a community we found at the Random Matrices and Applications workshop was vital. We are both entering the last year of our respective programs this fall and intend to continue working in academia. The relationships we established with the other attendees at the workshop will likely be strengthened for many years to come. As many in the research community know, even informal interactions can open the door for future collaborations.

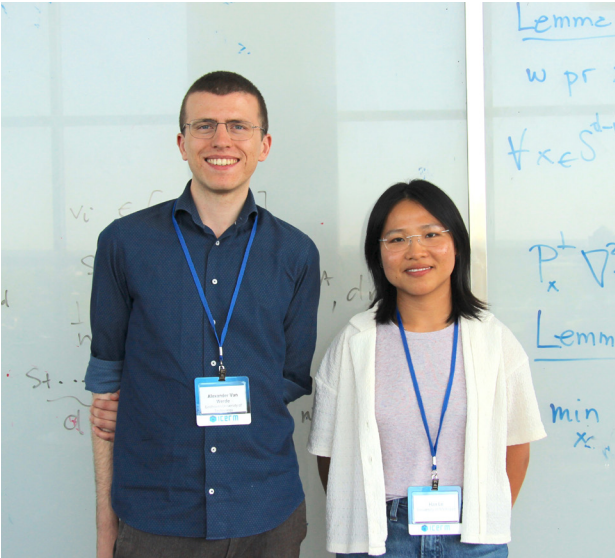
The chance to discuss our work with senior researchers was greatly appreciated. For instance, Alexander’s recent research developed random matrix-theoretic tools that can potentially be used to determine whether the isomorphism type of a random graph can be recovered from spectral information, an old open question in spectral graph theory. It was inspiring to discuss the new link with Hoi Nguyen and Sean O’Rourke, two professors at the workshop who had previously worked on related tools, although with totally different motivations.

As for Han, she had long discussions on Ising spin models with fellow graduate students from other institutions. She and her collaborators recently released [a paper on several variations of a spherical Sherrington–Kirkpatrick model](#), known as the spherical Sherrington–Kirkpatrick model, using eigenvalue results in random matrix theory to address the interaction matrix. Since they recently started focusing on problems in the Ising rather than the spherical case, being able to learn and

exchange ideas for the new direction during Random Matrices and Applications was a valuable experience.

One of the workshop’s most memorable presentations was [Shuangping Li’s talk](#) on the Ising perceptron model. The model originates from the theory of neural networks and concerns the intersection of $\{-1,1\}^n$ and many random half-spaces. Li’s talk addressed several thresholds

wherein there exists an efficient algorithm to sample points in the intersection set. Through diagrams describing multiple regimes where algorithmic results have either been proven or conjectured, Li drew a clear picture of the state of the problem, the present obstacles, and her contribution. It was also fascinating to hear Li’s story of how, during her travel for the workshop, she was able to improve on the very results she presented!

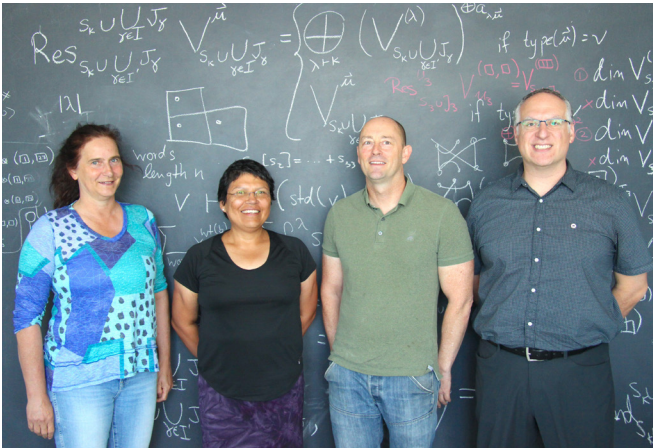


Van Werde (left) and Lee at ICERM

The proximity of ICERM to other hubs for scientific research, such as Boston and New York, made it easy to combine the workshop with other research visits. For instance, Alexander visited researchers at Princeton University and Harvard University the week before the Random Matrices conference. ICERM also provides a beautiful space that lends itself to collaboration—there are no difficulties when transitioning from lecture to coffee break where one can chat about research using the conveniently located whiteboards. And of course, we can’t forget the beautiful view from the eleventh floor! ■

A Computational Approach to Plethysm

From June 24-28, [Rosa Orellana](#) (Dartmouth College), [Franco Saliola](#) (Université du Québec à Montréal), [Anne Schilling](#) (University of California Davis), and [Mike Zabrocki](#) (York University) visited ICERM for the Collaborate program A Computational Approach to Plethysm. Bringing expertise in algebraic combinatorics, representation theory, and computation, the group gathered to continue their work on a classical problem in representation theory.



ICERM: Can you share a bit about the history that led up to this collaboration?

Zabrocki: We started this collaboration probably six years ago, trying to push forward a hundred-year-old problem in representation theory: How do you decompose a composition of representations into irreducibles? We've been moving forward for the last six years and it's going to take another six, but we can kind of see the light at the end of the tunnel. We really made progress this week, so things went well.

ICERM: What have you been working on this week?

Zabrocki: The main thing we tried to do is understand this representation formed by induction. So we had this awareness that the next steps we needed to do were compute these inductions within submonoids of this algebra we're looking at. And we sat down at the beginning of the week. Within two or three days, we had a good picture of what these objects look like. [We will] possibly be able to program them and compute good examples. That's the big step forward. And then to apply it to this other program, that's the next step. So I think in the past two days, we've really made progress connecting those things.

Schilling: I feel like we've tried to attack this problem from various angles for a long time and now we are using methods from semigroup theory which we hadn't really thought of. We are more trained in representation theory and combinatorics, so now we use methods from other areas to help us understand the problem and I feel like that's really helped us this week to make progress--this other angle.

Orellana: We were able to work examples that led to being able to program them, come up with ideas on how it would generalize and if this works we are very hopeful... This is a big problem we are going to solve, I feel, if all the ideas we generated work. Because as Anne mentioned, we have been looking at this from so many different dimensions that it will [unify] a lot of different things just solving this special case of the problem that we have identified at this point.

ICERM: What computational tools have you been using this week?

Saliola: We haven't technically been using ICERM's infrastructure but we have been using [SageMath] to explore a lot of the ideas to help us with the computations.

ICERM: What other research areas do you think will benefit from this collaboration?

Zabrocki: Just computation in general. I mean the whole idea is that linear algebra can be extremely slow when you implement it on a computer and here we're saying we can do these computations, transform

them into a set of small discrete objects, translate them back, and speed up the computation by one hundred-fold, in theory.

Orellana: The numbers that we are trying to compute are of much interest in computational complexity. If we are able to come up with an algorithm, I think immediately people will pick it up to try to understand the complexity of doing this problem in full. So I think it's going to have a lot of impact.

Zabrocki: Within the first couple of days, we had the mathematics we needed to probably write at least one paper. And then in the last couple days, three of our papers just kind of fit together like puzzle pieces and we've been able to make the connections that we didn't see ourselves much earlier in our project.

Orellana: Yeah, one of our papers is called ["The Mystery"](#) and the mystery is solved! ■

Mathematical Modeling, Deep Learning, and Optimization on Graph Structured Data

From May 13-17, [Weihong Guo](#) (Case Western Reserve University), [Yifei Lou](#) (University of North Carolina at Chapel Hill), [Jing Qin](#) (University of Kentucky), [Yangyang Xu](#) (Rensselaer Polytechnic Institute), and [Liangliang Zhang](#) (Case Western Reserve University) gathered for the hybrid Collaborate@ICERM program Mathematical Modeling, Deep Learning, and Optimization on Graph Structured Data. The group's research is motivated by applications in spatial transcriptomics, a cutting-edge molecular profiling method that allows biomedical researchers to map gene expression in complex tissues.



ICERM: Can you share a bit about your research topic and what you've been working on this week?

Zhang: Our modeling effort is motivated by a very popular application in biomedical research, which is called spatial transcriptomics. Spatial transcriptomics is crucial in medical research because it allows scientists to map gene expression directly with cell samples preserving the spatial context. This technology helps researchers understand the intricate relationship between cellular function and cell architecture, which is essential for studying complex diseases like cancer. By revealing how gene activity varies across different genes of a cell, spatial transcriptomics provides insights into the microenvironment's role in disease prevention and treatment response. This spatially-resolved gene expression data can lead to more accurate diagnoses, targeted therapies, and a deeper understanding of disease mechanisms at a cellular level. Because this data is very rich, it involves the graph and also high-dimensional data. It also requires us to classify the different gene types into the different cells. At the same time, we need to do a convolution to try to get the cell convolution at each location based on two types of data, so basically it's a data integration model.

Qin: During this week, we have explored developing a new mathematical model to analyze this special type of data and we tried to embed graph learning techniques, optimization techniques, and other data classifying algorithms, et cetera.

Xu: Yeah, we developed a solver – actually, two solvers – and we tested on some clean data. They work pretty [well] but now the challenge is doing this on real data. We're still working hard on that part.

Guo: Yeah, so the model has [a lot of] novelty as Liangliang was talking about. Spatial transcriptomics itself is a cutting-edge technology, so what we're doing is from the methodology aspect. So we're doing simultaneous cell and cell-type classification and deconvolution, so it's a very novel approach but it also brings in a lot of computational challenges. So I think this week has been very productive. We all have complimenting backgrounds and we are able to find the problem and talk it out and solve it, so thanks to ICERM for the support.

ICERM: What computational tools have you been using this week?

Guo: We are developing our own software but [ICERM's] IT support is very good!

ICERM: What other research areas do you think will benefit from this collaboration?

Zhang: Maybe in the future, because this method can benefit the area of biomedical research, we can apply this to cancer research [and try to] do a process prediction of the cancer progress...At the same time, we can continue to build the model and try to make it a platform like a website so that other people can benefit from our knowledge.

Xu: Our model is novel. Maybe it can apply to some other areas.

Guo: Yeah, it has a lot of connections with other applications. Like in remote-sensing, people do unmixing. Also in some other different applications in biology, for instance. Mathematical models are similar, so it has a lot of potential to be transformed to other areas and applications. ■

Early Graduate Researchers Grow Roots of Unity

In June, the screens in ICERM's lecture hall lit up with the words "inspired," "energized," and "healed." These were the words students used to describe their outlooks in a group poll at the end of *Roots of Unity*, a workshop for those early in their graduate studies in number theory, combinatorics, commutative algebra, geometry, and topology.

Organized by [Christine Berkesch](#) (University of Minnesota), [Michelle Manes](#) (American Institute of Mathematics), [Priyam Patel](#) (University of Utah), [Candice Price](#) (Smith College), and [Bianca Viray](#) (University of Washington), the biennial *Roots of Unity* workshop is tailored to support women and gender non-conforming students of color in mathematics. Specifically, it aims to address a gap in resources for underrepresented researchers between their undergraduate studies and dissertation work.

The idea for *Roots of Unity* came to the organizing committee on Blackout STEM Day, a strike for scientists organized by the #ShutDownSTEM movement amid the 2020 anti-racist protest

movement across the United States. On the day of the protest, arithmetic geometer and number theorist Bianca Viray invited a group of faculty to brainstorm ways to support and diversify their research communities. "We were thinking about things that we could do to affect our community in a positive direction and this workshop grew out of that conversation," said Christine Berkesch, a commutative algebraist and algebraic geometer.

"There are these kinds of pinch points in your career as a mathematician," explained Michelle Manes, a number theorist. The group asked themselves, "What are the things that supported you and got you through and may be missing for students from marginalized groups?"

Longstanding initiatives such as the [Women in Numbers](#) (WIN) research network and [Enhancing Diversity in Graduate Education](#) (EDGE) helped the organizers chart a path. Recognizing the efficacy of WIN and EDGE programs in supporting experienced researchers and recently admitted doctoral students, respectively, they decided to

create a workshop for Ph.D. candidates transitioning into independent research in their chosen fields. "We wanted to bridge these two things: one that maybe builds a support network earlier but not in your research area and this research [community] that comes much later," Viray explained.



Roots of Unity mentor Noelle Sawyer addresses members of the geometry and topology working group

The [first Roots of Unity](#) program was held at the University of Minnesota in 2022, before it was formally adopted by the [National Science Foundation Mathematical Sciences Institutes Diversity Initiative](#). Its success laid the foundation for the 2024 program, where small groups of students spent the week working with early-career and faculty mentors to break down complex papers in their fields. Program mentors [Renee Bell](#), [Bianca Thompson](#), [Noelle Sawyer](#), [Chandrika Sadanand](#), [Haydee Lindo](#), [Ashley Wheeler](#), [Naiomi Cameron](#), and [Shanise Walker](#) passed down research tips like keeping a math journal, reading in groups, and digging into a conjecture's history to understand it better. Between working group sessions, participants spent time in professional development and belonging seminars, as well as touring the American Mathematical Society's local headquarters and printing facilities.

"I actually shared the finding support systems slides with some of my friends afterwards because I just found them super useful," said [Judy Chiang](#), a second year mathematics Ph.D. student at the University of Minnesota and a member of the commutative algebra working group. According to Chiang, the workshop struck an ideal balance of mathematical work, community building, and professional development. "Mathematically, I

learned more about [our topic] and how to approach the math when I'm reading literature. Socially, I feel [that] these peers are going to become my academic friends and we can study together."

New friendships and the program's highly collaborative research culture were on display during students' final presentations, where every working group gave a collaborative talk on their assigned paper. Students demonstrated what they learned about monodromy and Galois groups, Gorenstein rings, combinatorial problems in Dyck paths and parking functions, and universal covers of manifolds. Each presentation was enthusiastically praised by participants, program leaders, and ICERM Associate Director for Diversity and Outreach Ulrica Wilson.

On the organizers' part, the workshop provided an opportunity to develop a guide in hopes of inspiring similar events in other research areas. "We're not going to do applied math or analysis or PDEs, but we would love for other people to do that. So we now have a template that we could hand off to organizers who wanted to think about doing something similar in their research areas," Manes said. "It's about capacity and ability," Viray added. "If I meet someone in analysis, I can try to give them all the great advice I can but I can't advocate for them in the same way or really help them think about their research proposal the same way I can in number theory."

The organizing committee is also determined to foster an active network of *Roots of Unity* alumni. Their goals, explained Berkesch, are "to grow the group and keep the network around to stay with the students so they can grow together."

Thinking far into the future, Manes shared her hopes for this year's *Roots of Unity* participants. "I would love to see in ten years from now that there's an impact, that the fields of number theory, combinatorics and topology look different at the faculty level because these students have been supported in these transition points." ■

The organizing committee of Roots of Unity 2024 would like to acknowledge Adriana Salerno for her contributions to the program.